

```

from itertools import permutations

d = [[0, 10, 15, 20],
      [10, 0, 35, 25],
      [15, 35, 0, 30],
      [20, 25, 30, 0]]

cost = 999

path = ()

for p in permutations([1, 2, 3]):
    c = d[0][p[0]] + d[p[0]][p[1]] + d[p[1]][p[2]] + d[p[2]][0]
    if c < cost:
        cost = c
        path = (0,) + p + (0,)

print("Shortest Path:", path)
print("Minimum Cost:", cost)

```

OUTPUT:

```

Shortest Path: (0, 1, 3, 2, 0)
Minimum Cost: 80

```